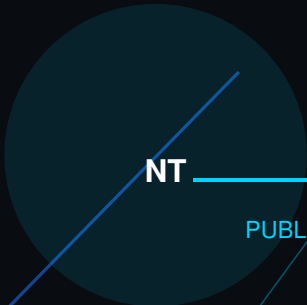




Node: The Building Block of NervTech Solutions

Initial Project Overview

A plain-language briefing for stakeholders, collaborators, and portfolio reviewers

PROJECT

Node v0

Document ID	NT-N-OVR-001	Revision	A
Status	Draft Brief	Date	2026-06-25
Audience	Non-technical stakeholders; Portfolio reviewers	Owner	NervTech Systems Engineering
Author	Anhadh Virk	Class	PUBLIC

Purpose

To pave the way for NervTech robotic systems, a robust, multipurpose actuator must be developed that has high precision, bandwidth and torque capabilities. The Node v0 fills this need, while being developed in house without the extra integration time and cost incurred by OEM integrated actuators.

Summary

This overview is intentionally non-technical. It covers the current stakeholder need for the product, how it fits into the growth of NervTech, the requirements that define success for the product and finally a roadmap to achieve this success. It avoids equations and low-level implementation detail so the project can be understood by reviewers, collaborators, mentors, and potential employers.

CONTENTS

1	Project Snapshot	2
2	What the Project Is	2
3	Why It Matters	2
4	Who This Is For	3
5	Current Status	3
6	Roadmap	3
7	Communication Positioning	4

1 PROJECT SNAPSHOT

Project Snapshot

What it is	A compact robotic actuator module that combines motion generation, sensing, electronics, and control in one reusable unit.
Why it matters	Robots become easier to build when the joint actuator is reliable, measurable, and simple to integrate.
Current stage	The project is in early architecture and prototype planning.
Next visible result	A bench-tested actuator module with plots, photos, CAD renders, and a short demonstration video.

2 WHAT THE PROJECT IS

Overview – Plain-Language Summary

The Integrated Robotic Actuator is a self-contained robotic joint module. Instead of treating the motor, gearbox, sensors, electronics, and software as separate parts, the project brings them together into one engineered actuator package.

The goal is not only to make something move. The goal is to show a complete robotics engineering workflow: research, design, prototyping, testing, documentation, and demonstration.

3 WHY IT MATTERS

Value – Project Value

A well-designed actuator is a building block for many robotic systems, including arms, mobile robots, legged robots, grippers, and educational robotics platforms.

For a portfolio, this project is valuable because it connects mechanical design, electronics, embedded control, safety testing, data collection, and clear communication.

4 WHO THIS IS FOR

Table 1: Stakeholder Map

Audience	What they need to know	How this document helps
Engineering reviewers	Whether the project has credible technical depth.	Shows architecture, testing intent, and visible deliverables.
Research mentors	Whether the project can connect to control, learning, or neuromorphic sensing.	Frames the project as a reusable experimental platform.
Non-technical stakeholders	What the project does and why it matters.	Explains the project without relying on formulas or jargon.

5 CURRENT STATUS

Status – Now

The project is at the point where the major design direction should be selected, documented, and validated through a small bench prototype.

Table 2: Milestones

Milestone	Meaning	Target	Status
Concept	Define the actuator purpose and intended demonstration.	Current	In progress
Architecture	Choose motor, sensing, gearbox, controller, and package strategy.	Next	Planned
Prototype	Build and test a first integrated bench unit.	Following	Planned
Portfolio	Publish video, plots, CAD renders, README, and lessons learned.	Final	Planned

6 ROADMAP

Table 3: Project Roadmap

Phase	Focus	Visible Output
Phase 1	Research and concept selection.	Preliminary research note
Phase 2	Technical design and requirements.	Engineering design document
Phase 3	Prototype build and bench testing.	Photos, plots, test logs
Phase 4	Portfolio packaging and public explanation.	GitHub README and demo video

7 COMMUNICATION POSITIONING

Stakeholder Note – How to Explain It

NervTech is building a compact actuator module that can become the motion core of future robotic systems. The project demonstrates not just a part, but an engineering process: make a design decision, build it, test it, learn from it, and document the result clearly.

Concern – Communication Risk

Avoid describing the project as a finished product too early. Present it as a serious prototype and learning platform until the actuator has been built and tested.